ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE C · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mechatronics paper 1 · practice c.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and examples.

SECTION A

QUESTION 1

1.1 1.1.1 A ✓ 1.1.2 B ✓ 1.1.3 C ✓ 1.1.4 B ✓ 1.1.5 A ✓ 1.1.6 C ✓ 1.1.7 B ✓ 1.1.8 D ✓ 1.1.9 A ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 False ✓ 1.2.4 True ✓ 1.2.5 False ✓ 1.2.6 False ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 True ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 C ✓ 1.3.2 G ✓ 1.3.3 I ✓ 1.3.4 B ✓ 1.3.5 E ✓ 1.3.6 J ✓ 1.3.7 A ✓ 1.3.8 F ✓ 1.3.9 D ✓ 1.3.10 H ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 The combination of mechanical engineering, electronics, control engineering and computing ✓ to design smart products and systems that sense, decide and act ✓. (2)

2.2 Sensors – detect walls, dirt, cliffs (bump / IR sensors) ✓; signal conditioning – changes sensor signals into a usable form (amplify, filter, convert to digital) ✓; controller – microcontroller decides what to do ✓; actuators – motors drive the wheels, brushes and suction fan ✓. (4)

2.3 Any TWO ✓✓: automatic washing machine, ATM, car ABS / engine management, digital camera, automatic gate motor, vending machine, CNC machine, 3D printer, industrial robot. (2)

2.4.1 $\mathrm{V} = 2\ \mathrm{mV} \times 85\ ✓ = 170\ \mathrm{mV}\ ✓$ (2)

2.4.2 $\mathrm{Output\ at\ 150\ °C} = 2 \times 150 = 300\ \mathrm{mV}\ ✓; \mathrm{span} = 300 - 0 = 300\ \mathrm{mV}\ ✓$ (2)

2.5 Range: the lowest to the highest input the sensor can measure ✓. Resolution: the smallest change in input that the sensor can detect ✓. (2)

[14]

QUESTION 3

3.1 A current in the coil creates a magnetic field ✓ that pulls a soft-iron plunger into the coil ✓; when the current stops, a spring pushes the plunger back ✓. (3)

3.2.1 $\mathrm{Steps} = 90 \div 1{,}8\ ✓ = 50\ \mathrm{steps}\ ✓$ (2)

3.2.2 $\text{Steps per revolution} = 360 \div 1{,}8 = 200$ ✓; $\text{rev/s} = 400 \div 200 = 2$ ✓; $\text{N} = 2 \times 60 = 120\ \text{r/min}$ ✓ (3)

3.3 The servo's controller compares the pulse width with the position measured by its built-in potentiometer and turns until they match ✓; a longer pulse gives a larger angle ✓. 1,5 ms → 90° ✓ (3)

3.4 One advantage ✓ and one disadvantage ✓ each: electric – precise control, clean, easy to connect / lower force, can overheat; pneumatic – fast, cheap, clean, safe in explosive areas / hard to stop at exact positions, noisy, needs a compressor; hydraulic – very large forces, precise and stiff / leaks, oil is messy and a fire risk, expensive. (6)

3.5 Any ONE ✓: lifting a platform, pushing parts off a conveyor, opening a gripper, adjusting a solar panel, moving a 3D-printer or CNC axis. (1)

[18]

QUESTION 4

4.1.1 $\text{A} = \pi\text{D}^2 \div 4 = \pi \times 0{,}05^2 \div 4 = 1{,}96 \times 10^{-3}\ \text{m}^2$ ✓; $\text{F} = \text{P} \times \text{A}$ ✓ $= 600\,000 \times 1{,}96 \times 10^{-3} = 1\,178{,}1\ \text{N}$ ✓ (3)

4.1.2 $\text{A} = \pi(\text{D}^2 - \text{d}^2) \div 4 = \pi(0{,}05^2 - 0{,}02^2) \div 4 = 1{,}65 \times 10^{-3}\ \text{m}^2$ ✓; $\text{F} = 600\,000 \times 1{,}65 \times 10^{-3}$ ✓ $= 989{,}6\ \text{N}$ ✓ (3)

4.1.3 The piston rod takes up part of the piston area on the retracting side, so the air pushes on a smaller area ✓ (1)

4.2 The first number is the number of ports and the second the number of positions ✓. 5/2: five ports, two positions ✓ – controls a double-acting cylinder ✓; 3/2: three ports, two positions – controls a single-acting cylinder ✓. (4)

4.3 Filter – removes dirt and water from the air ✓; regulator – keeps the air pressure at a set, steady value ✓; lubricator – adds a fine oil mist to lubricate valves and cylinders ✓. (3)

4.4 $\text{F}_2 = \text{F}_1 \times \text{A}_2 \div \text{A}_1$ ✓ $= 100 \times 50 \div 2$ ✓ $= 2\,500\ \text{N}$ ✓ (3)

4.5 Any ONE ✓: never point an air gun at anyone; wear safety glasses; switch off and release the pressure before disconnecting pipes; check pipes and fittings for damage. (1)

[18]

QUESTION 5

5.1.1 $\text{GR} = 60 \div 20 = 3:1$ ✓ (1)

5.1.2 $\text{N}_2 = \text{N}_1 \div \text{GR}$ ✓ $= 1\,500 \div 3 = 500\ \text{r/min}$ ✓ (2)

5.1.3 $\text{T}_{\text{out}} = \text{T}_{\text{in}} \times \text{GR} \times \eta$ ✓ $= 4 \times 3 \times 0{,}95$ ✓ $= 11{,}4\ \text{N·m}$ ✓ (3)

5.2.1 $\text{N}_2 = \text{N}_1 \times \text{D}_1 \div \text{D}_2$ ✓ $= 1\,440 \times 100 \div 240 = 600\ \text{r/min}$ ✓ (2)

5.2.2 $\text{v} = \pi\text{DN} \div 60$ ✓ $= \pi \times 0{,}1 \times 1\,440 \div 60$ ✓ $= 7{,}54\ \text{m/s}$ ✓ (3)

5.3 Any TWO differences ✓✓✓✓: a chain drive does not slip, a belt can slip; chains need lubrication, belts do not; belts are quieter and absorb shock; chains carry larger loads at lower speeds; belts can run at higher speeds and over longer distances; chains last longer in hot or oily places. (4)

5.4 Any THREE ✓✓✓: preventive – planned at fixed times (oil changes, inspections); predictive / condition-based – based on measurements (vibration, temperature, oil analysis); breakdown / corrective – repair after a failure. (3)

5.5 Any TWO ✓✓: reduces friction and wear; carries heat away (cooling); prevents rust / corrosion; keeps out dirt and seals; reduces noise. (2)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE